

CHAPTER 1

INTRODUCTION

In modern power systems, the rapid increase in nonlinear loads, particularly Electric Vehicle charging stations, has led to significant power quality issues in three-phase systems. These loads draw non-sinusoidal currents, resulting in harmonics, increased reactive power demand, and voltage disturbances in the electrical network. Such effects degrade the overall system performance by increasing Total Harmonic Distortion, reducing power factor, and causing inefficiencies and potential damage to electrical equipment. Therefore, maintaining power quality within acceptable limits has become a crucial aspect of power system operation, especially with the growing adoption of EV infrastructure and smart grid technologies.

1.1. Rationale

The increasing adoption of Electric Vehicle charging systems has introduced significant power quality challenges in three-phase power systems due to their nonlinear nature. These loads generate harmonics, increase reactive power demand, and cause voltage disturbances, which result in poor power factor, increased Total Harmonic Distortion, and reduced system efficiency. Conventional methods are often insufficient to address these issues effectively, especially when multiple power quality problems occur simultaneously. Therefore, there is a need for an advanced and integrated solution that can both monitor and mitigate these disturbances in real time.

This project is motivated by the need to develop a reliable system that not only analyzes power quality parameters but also enhances them using an efficient compensation technique. By implementing a hybrid system combining a Shunt Active Power Filter and a capacitor bank, the project aims to reduce harmonics and provide reactive power compensation simultaneously. This approach ensures improved current quality, better voltage profile, and enhanced overall system performance, making it highly suitable for modern applications such as EV charging stations and smart grid systems.

CHAPTER 2

LITERATURE SURVEY

[1] “Harmonic Mitigation and Energy Savings using P-Q Based Shunt Active Filter” – M. S. Arjun et al., 2025

This paper focuses on the application of a P-Q theory-based Shunt Active Power Filter for reducing harmonic distortion in power systems. The authors designed and implemented an active filtering technique that dynamically compensates harmonic currents generated by nonlinear loads. Their work demonstrated that the SAPF effectively improves power quality by reducing Total Harmonic Distortion and enhancing energy efficiency. The study highlights the capability of active filters in maintaining sinusoidal source currents and improving overall system performance under distorted load conditions.

[2] “Hybrid Power Filter for Harmonic Reduction” – Waheed et al., 2024

In this research, the authors proposed a hybrid power filter combining active and passive filtering techniques to mitigate harmonics in electrical systems. The study showed a significant reduction in THD from 29.85% to 0.78%, proving the effectiveness of the hybrid approach. The passive filter helps in handling lower-order harmonics, while the active filter compensates higher-order harmonics dynamically. The results indicate improved voltage quality, better power factor, and enhanced system stability, making hybrid filters a reliable solution for modern power quality issues.

[3] “Impact of Electric Vehicle Charging Station on Power Quality” – Condemaita

This paper investigates the effect of Electric Vehicle charging stations on the power quality of three-phase systems. The authors analyzed how EV chargers, acting as nonlinear loads, introduce harmonics, increase reactive power demand, and distort voltage and current waveforms. The study applied advanced techniques such as P-Q theory and Synchronous Reference Frame methods to mitigate these issues. The results showed that THD was reduced from 28.24% to 6.23% using P-Q theory and further down to 2.92% using SRF, demonstrating the importance of effective compensation methods in EV-integrated power systems.

CHAPTER 3

SYSTEM REQUIREMENTS

The system requires proper integration of measurement, control, and compensation components to analyze and improve power quality in a three-phase system. It includes both software and functional capabilities to handle nonlinear loads and enhance system performance effectively.

3.1. Software Requirements

- The system requires MATLAB/Simulink software to model and simulate the three phase power system and nonlinear EV load.
- The software is used to design and implement the Shunt Active Power Filter control algorithm.
- The software provides waveform visualization for better understanding of system performance before and after compensation.
- The programming environment helps in tuning and optimizing the filter parameters for better performance.

3.2. Functional Requirements

- The system must measure electrical parameters such as voltage, current, active power, reactive power, and THD at the Point of Common Coupling .
- The system should detect harmonic distortions caused by the nonlinear EV charging load.
- The system must implement a Shunt Active Power Filter to compensate harmonic currents dynamically.
- The system should display waveform outputs and parameter values for analysis and verification.

- The system must ensure proper synchronization with the three-phase supply for accurate operation.

3.3. Non-Functional Requirement

The system should provide high accuracy in measuring electrical parameters to ensure reliable results.

- The system must operate in real-time to detect and compensate harmonics without delay.
- The system should maintain stability under varying load conditions, especially during EV charging.
- The system must be reliable for continuous operation without frequent failures.
- The system must be efficient in operation with minimal power losses during filtering and compensation.
- The system should have a user-friendly interface for easy monitoring and control.
- The system should ensure safety and proper isolation while handling high voltage and current levels.

3.4. System Constraints

- The system is limited by the availability and accuracy of sensors used for measurement.
- The system performance depends on proper calibration of instruments and synchronization of signals.
- The design and tuning of the Shunt Active Power Filter require complex control strategies.
- The implementation cost may restrict the use of advanced filtering components in practical systems.
- The system may be affected by external disturbances and environmental conditions.

CHAPTER 4

METHODOLOGY

The proposed work focuses on analyzing and improving power quality in a three-phase system supplying a nonlinear EV charging load using a hybrid compensation technique. The methodology involves systematic modeling, measurement, analysis, and implementation of filtering techniques to reduce harmonics and improve system performance.

1. System Modeling of Three-Phase Supply

In this stage, a three-phase power system is modeled using MATLAB/Simulink to represent real-time grid conditions. The system is designed with standard voltage and frequency values, and all components are interconnected at the Point of Common Coupling , where measurement and compensation take place.

2. Nonlinear EV Load Implementation

A nonlinear load representing an EV charging station is implemented using a diode rectifier with an RL load. This load draws non-sinusoidal current from the source, which introduces harmonics, increases reactive power demand, and degrades overall power quality.

3. Power Quality Measurement at PCC

A power quality analyzer is designed to measure key electrical parameters such as RMS voltage, RMS current, active power, reactive power, apparent power, power factor, and Total Harmonic Distortion . These measurements are taken at the PCC to evaluate the impact of the nonlinear load on the system.

4. Power Quality Assessment without Compensation

The system is initially analyzed without any compensation techniques to observe the extent of power quality degradation. Parameters such as high THD, poor power factor, and distorted waveforms are identified, which justify the need for improvement.

5. Design and Implementation of Shunt Active Power Filter

A Shunt Active Power Filter is implemented using an IGBT-based Voltage Source Inverter connected in shunt at the PCC. The SAPF operates by injecting compensating current equal and opposite to harmonic components, thereby improving current waveform quality.

6. Reference Current Generation using Control Techniques

The SAPF control system uses a Phase Locked Loop to synchronize with the grid frequency. The three-phase signals are transformed into dq0 reference frame using abc–dq0 transformation, and a low-pass filter is used to separate fundamental and harmonic components, generating the reference compensation current.

7. Hysteresis Current Control for Switching

A hysteresis current controller is used to compare the actual current with the reference current and generate gate pulses for the IGBT inverter. This ensures fast and accurate tracking of the compensation current for effective harmonic elimination.

8. Reactive Power Compensation using Capacitor Bank

A three-stage capacitor bank is implemented to provide reactive power compensation. The capacitors are automatically switched based on power factor thresholds (such as 0.95, 0.90, and 0.85), which helps in improving power factor and reducing reactive power demand.

9. Hybrid Compensation System Integration

The SAPF and capacitor bank are combined to form a hybrid compensation system. The capacitor bank handles bulk reactive power, while the SAPF eliminates dynamic harmonics, resulting in efficient and cost-effective power quality improvement.

10. Performance Evaluation and Comparison

Finally, the system performance is evaluated by comparing parameters before and after compensation. The results show a significant reduction in THD and improvement in power factor, achieving values within IEEE 519 standards and enhancing overall system efficiency.

CHAPTER 5

IMPLEMENTATION

The system is implemented using MATLAB/Simulink by modeling the three-phase supply, nonlinear EV load, and hybrid compensation system. The SAPF and capacitor bank are integrated at the PCC to reduce harmonics and improve power factor.

5.1 Block Diagram of Power Quality Measurement and Enhancement using Hybrid Compensation Method

The flow diagram represents the step-by-step process of measuring, analyzing, and enhancing power quality using hybrid compensation.

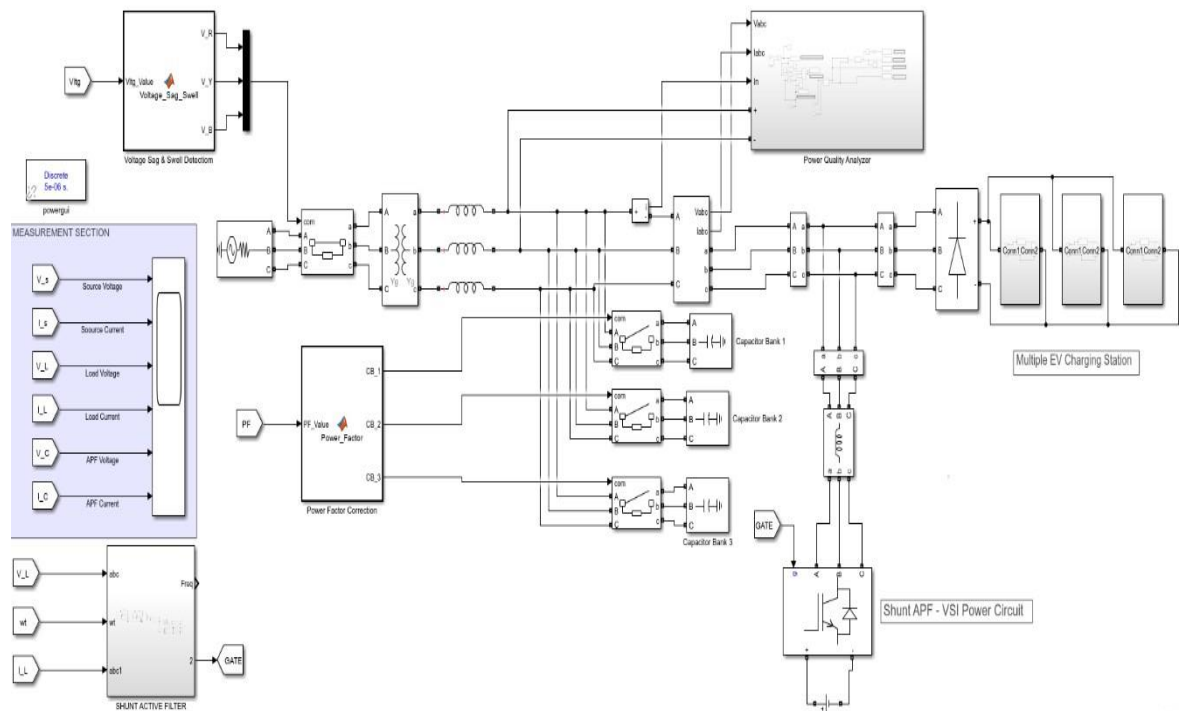
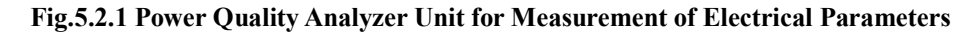


Fig. 5.1.1 Fig. Overall Block Diagram of Hybrid Compensation System for Power Quality Improvement in Three-Phase System

5.2 Power Quality Analyzer Unit

The power quality analyzer unit is used to measure key parameters such as voltage, current, power factor, and Total Harmonic Distortion at the Point of Common Coupling.



The SAPF unit is used to eliminate harmonic currents by injecting compensating current into the system. It improves current waveform quality and reduces THD in the three-phase system.



CHAPTER 6

RESULTS AND OUTPUT

The results show that the SAPF successfully minimizes harmonics and enhances power factor. The system achieves improved power quality and stable operation as verified through waveform analysis.

6.1 Performance Analysis of System Parameters

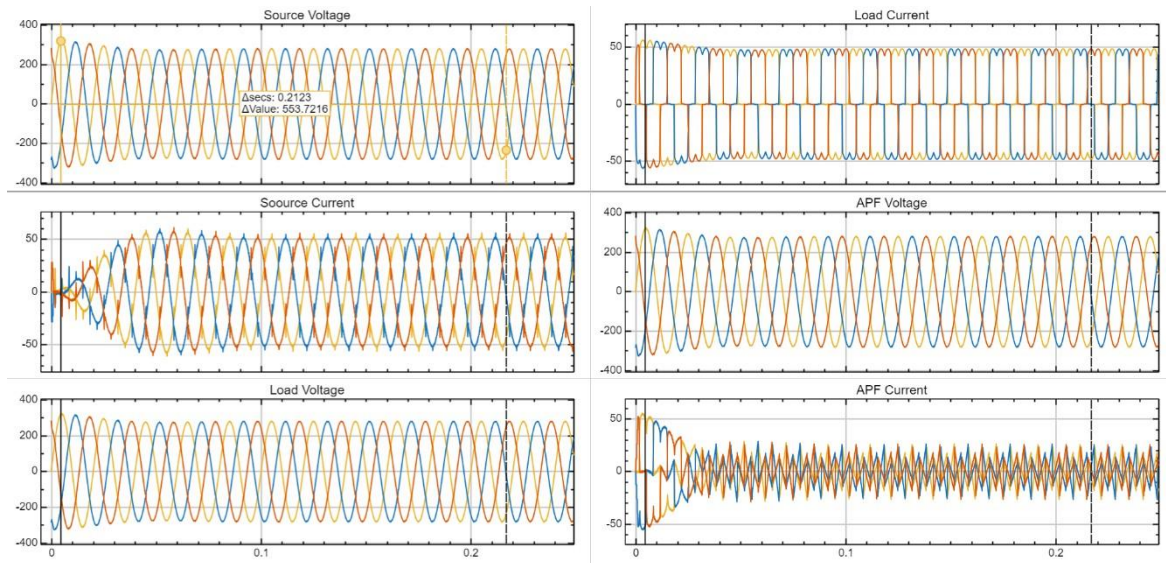


Fig.6.1.1 Combined Waveforms of Input Power, Output Power, and Source Voltage

This graph represents the variation of input power, output power, and source voltage with respect to time. It is used to analyze the overall performance and dynamic behavior of the system.

6.2 Harmonic Analysis Before Compensation

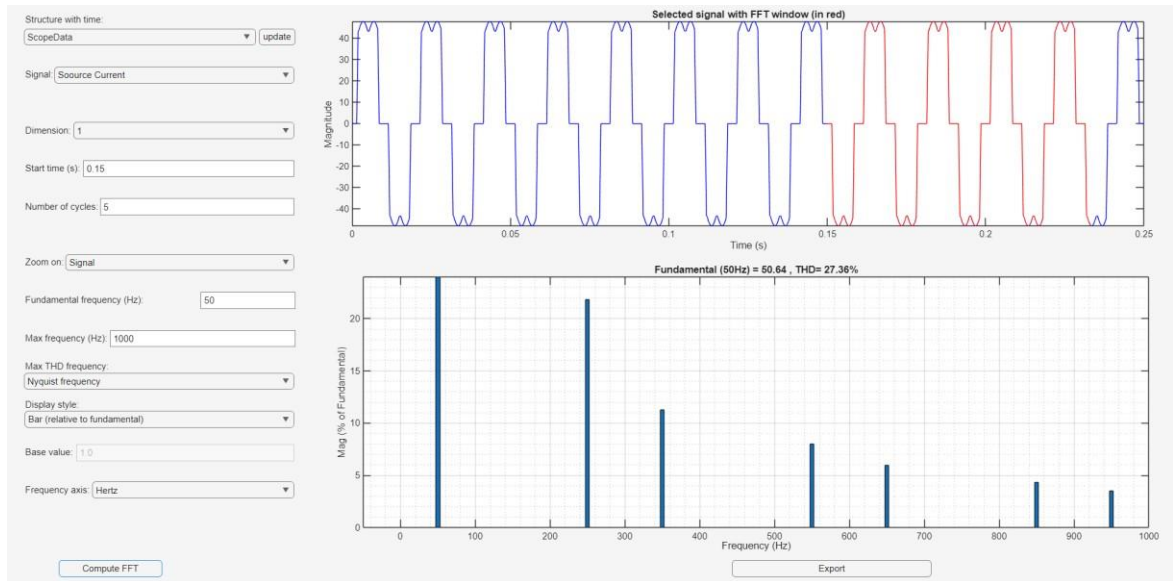


Fig.6.2.1 Total Harmonic Distortion (THD) Before Compensation

This figure shows the presence of significant harmonic distortion in the system before compensation. The waveform is non-sinusoidal, indicating poor power quality and the need for harmonic mitigation.

6.3 Harmonic Analysis After Compensation

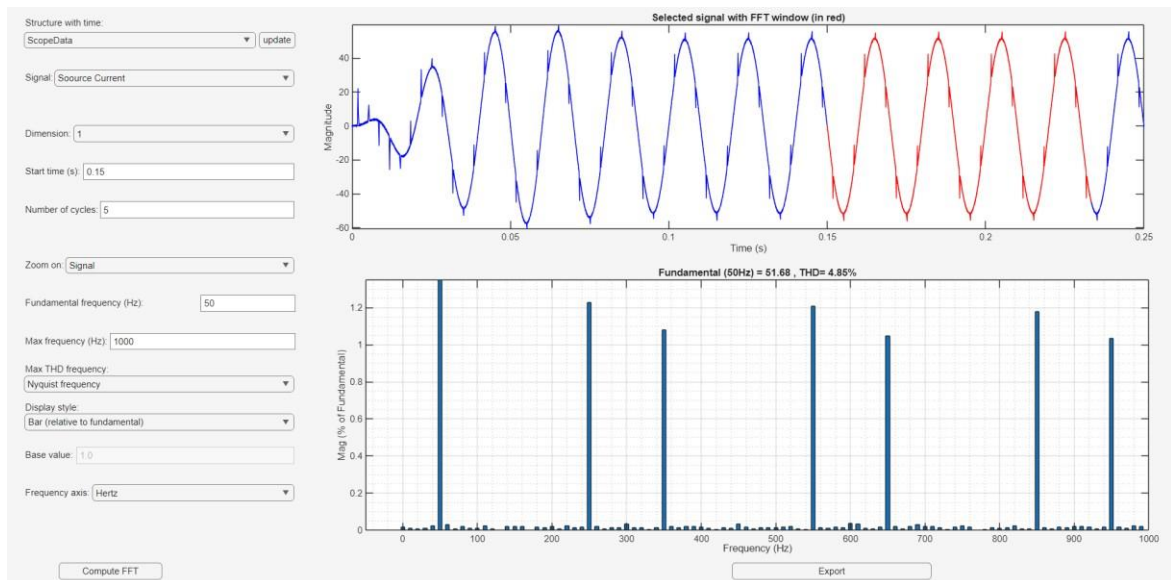


Fig.6.3.1 Total Harmonic Distortion (THD) After Compensation

This figure illustrates the reduction in harmonic distortion after the application of the Shunt Active Power Filter. The waveform becomes more sinusoidal, indicating improved power quality and enhanced system performance.

6.4 Comparison of Practical Values and Theoretical Values before compensation

Parameters	Practical Values	Theoretical Value
Voltage	415V	415V
Current	9.67A	9.67A
Power Factor	0.78	0.78
Apparent Power	6.98KVA	7.2KVA
Active Power	5.1KW	5.4KW
Reactive Power	4.32KVAR	4.76KVAR
THD	27.36%	27.36%

Table 6.4.1 Comparison before compensation

Theoretical Equations.

1. Apparent Power (S)

Formula:

$$S = \sqrt{3} \times V \times I$$

$$S = \sqrt{3} \times 415 \times 9.67$$

$$S \approx 7.2 \text{ KVA}$$

2. Active Power (P)

Formula:

$$P = \sqrt{3} \times V \times I \times \cos \phi$$

$$P = \sqrt{3} \times 415 \times 9.67 \times 0.78$$

$$P \approx 5.4 \text{ kW}$$

3. Reactive Power (Q)

Formula:

$$Q = \sqrt{3} \times V \times I \sin \phi$$

$$\sin \phi = \sqrt{1 - (0.78)^2} \approx 0.625$$

$$Q = \sqrt{3} \times 415 \times 9.67 \times 0.625$$

$$Q \approx 4.76 \text{ KVAR}$$

4. Power Factor Check:

$$\cos \phi = \frac{P}{S}$$

$$\cos \phi = \frac{5.4}{7.2} \approx 0.78$$

5. Total Harmonic Distortion (THD)

$$THD = \frac{\sqrt{(I_{rms})^2 - (I_1)^2}}{I_1}$$

$$I_{rms} = 10 \text{ A}$$

$$I_1 \approx 9.65 \text{ A}$$

$$THD = \frac{\sqrt{(10)^2 - (9.65)^2}}{9.65}$$

$$THD = \frac{\sqrt{100 - 93.12}}{9.65}$$

$$THD = \frac{\sqrt{6.87}}{9.65}$$

$$THD = \frac{2.62}{9.65}$$

$$THD \approx 0.2736$$

$$THD \approx 27.36\%$$

Fig. 6.4.1 Theoretical Calculation before compensation

6.5 Comparison of Practical Values and Theoretical Values After compensation

Parameters	Practical Values	Theoretical Value
Voltage	415V	415V
Current	9A	9A
Power Factor	0.95	0.95
Apparent Power	6.87KVA	6.76KVA
Active Power	5.67KW	5.88KW
Reactive Power	0.39KVAR	0.33KVAR
THD	4.8%	4.1%

Table 6.5.1 Comparison After compensation

Theoretical Calculations: (After compensating methods adding)

- * Voltage = 415V.
- * Current is reduced to 9.67 to 9A.
= 9A.
- * Power factor = 0.95.

→ Apparent power (S) ⇒ Total harmonic distortion (THD)

$$S = \sqrt{3} \times V \times I$$

$$= \sqrt{3} \times 415 \times 9.$$

$$S = 6.76 \text{ KVA.}$$

→ Active Power (P)

$$P = S \times \text{PF}$$

$$= 6.76 \times 0.95$$

$$= 5.88 \text{ KW}$$

→ Reactive Power (Q)

$$Q = \sqrt{S^2 - P^2}$$

$$Q \approx 0.33 \text{ KVAR.}$$

$$= \frac{\sqrt{I_3^2 + I_5^2 + I_7^2}}{I_1}.$$

$$I_1 = 9 \text{ A.}$$

$$I_3 = 0.3 \text{ A}$$

$$I_5 = 0.2 \text{ A}$$

$$I_7 = 0.1 \text{ A}$$

$$= \frac{\sqrt{0.14}}{9} = \frac{0.374}{9} \times 100$$

$$\text{THD} \approx 4.15\%.$$

Fig. 6.5.1 Theoretical Calculation After compensation

CHAPTER 7

APPLICATIONS

- **Electric Vehicle Charging Stations:** The system improves power quality by reducing harmonics caused by nonlinear EV loads.
- **Industrial Power Systems:** It helps industries maintain stable operation by minimizing harmonic distortion and improving power factor.
- **Smart Grid Systems:** The system supports advanced grid infrastructure by ensuring efficient and reliable power delivery.
- **Renewable Energy Integration:** It enhances power quality in systems integrated with solar and wind energy sources.
- **Commercial Buildings:** The system improves efficiency and reduces losses in buildings with heavy electrical loads.
- **Data Centers:** It ensures stable and distortion-free power supply for sensitive electronic equipment.
- **Distribution Networks:** The system maintains voltage profile and reduces disturbances in power distribution systems.

CHAPTER 8

CONCLUSION

The project successfully demonstrates the measurement and enhancement of power quality in a three-phase system using a hybrid compensation method. The presence of nonlinear loads such as EV charging systems introduces harmonics, reactive power demand, and voltage disturbances, which significantly degrade system performance. By analyzing these effects, the necessity for effective power quality improvement techniques is clearly established.

The implementation of a Shunt Active Power Filter along with a capacitor bank provides an efficient solution to mitigate these issues. The SAPF effectively eliminates harmonic currents using advanced control techniques such as PLL, abc–dq0 transformation, and hysteresis control, while the capacitor bank compensates reactive power and improves power factor. The integration of these components results in a hybrid system capable of handling both harmonic and reactive power problems.

The results obtained from the system clearly show a significant improvement in power quality parameters. The Total Harmonic Distortion is reduced from a high value (around 27%) to within IEEE 519 limits (around 4.8%), and the power factor is improved close to unity. These improvements validate the effectiveness of the proposed hybrid compensation technique.

Furthermore, the comparison between theoretical and practical results confirms the reliability and accuracy of the system design. The system operates efficiently under different load conditions and ensures stable performance, making it suitable for real-world applications.

Overall, the project provides a cost-effective and reliable solution for improving power quality in modern power systems, especially in applications involving nonlinear loads such as EV charging stations and smart grids.

CHAPTER 9

FUTURE ENHANCEMENTS

The proposed system can be further enhanced by incorporating advanced technologies and improving system capabilities for better performance and adaptability.

In future work, real-time hardware implementation of the system can be carried out using digital controllers such as DSP or FPGA. This will enable practical validation of the proposed model and allow the system to be deployed in real-world applications such as EV charging stations and industrial systems.

Advanced control algorithms such as artificial intelligence and machine learning techniques can be integrated to improve the performance of the SAPF. These techniques can enable adaptive control, faster response, and better handling of dynamic load variations in the system.

The system can also be extended to support renewable energy integration, such as solar and wind power systems. By combining hybrid compensation with renewable sources, the overall efficiency and sustainability of the power system can be improved.

Further improvements can be made by optimizing the design of the capacitor bank and inverter to reduce cost and improve efficiency. Advanced switching techniques can also be implemented to minimize losses and enhance system reliability.

Additionally, the system can be integrated into smart grid infrastructure with IoT-based monitoring and control. This will allow remote monitoring, real-time data analysis, and intelligent decision-making for improved power quality management.

REFERENCES

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